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Module I

Introduction, what is Artificial Intelligence (AI)?
Foundation of AI, The history of AI, Appln of AI
Intelligent agents - Agents and environments,
Good Behavior: The concept of rationality, The nature
of environments, The Structure of Agents.

Introduction

We call ourselves Homo Sapiens - Man the wise -
Because our Intelligence is so important to us.

For thousands of years, we have tried to understand
how we think; that is how a mere handful of
matter can perceive, understand, predict, and
manipulate a world far larger and more complicated
than itself. The field of artificial intelligence, or AI
goes further still; it attempts not just to understand
but also to build intelligent entities.

AI is one of the newest fields in science
and engineering. Work started in earnest soon -
after World War II, and the name itself was
coined in 1956. Along with molecular biology.

AI is regularly cited as the "field" would most
like to be in "by scientists in other disciplines.

AI is currently encompasses a huge variety
of subfields, ranging from the general
(learning and perception) to the specific, such

as playing chess, solving Mathematical theorem,
writing poetry, driving a car on a crowded
street, and diagnosing diseases. AI is relevant
to any intelligent task; It is truly a universal
field.

Father of AI $\rightarrow$ John McCarthy (1956)

"The Science and Engineering of making Intelligent
machines".

Turing Test $\Rightarrow$ Alan Turing (In ability to distinguish

Computer responses from human
responses)

AI deals with helping machines find solutions
to complex problems in a more human like

fashion.

$\rightarrow$ Borrowing characteristics from human Intelligence
and applying them as algorithms in a computer

friendly way.

$\rightarrow$ NLP $\Rightarrow$ (Natural language processing) $\leftarrow$ Applⁿ.

- What is AI -

Based on Fig 1(a)

The definitions on top are concerned with thought processes and reasoning, whereas the ones on the bottom address behavior. The definitions on the left measure success in terms of fidelity to human performance, whereas the ones on the right measure against an ideal performance measure called - rationality. A system is rational if it does the "right thinking" "right thing" given what it knows.

low of thought approach → right thinking

Thinking Humanly

"The exciting new effort to make computers think... machines with minds, in the full and literal sense."

"The automation of activities that we associate with human thinking; activities such as decision making, problem solving, learning. (Bellman 1978)"

Thinking Rationally

The study of mental faculties through the use of computational models."

The study of computations that make it possible to perceive, reason, and act"

Acting humanly :-

The art of creating machines that perform functions that require intelligence when performed by people."

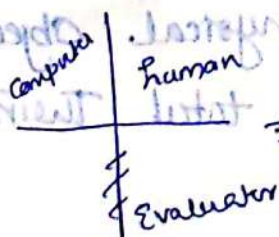
The study of how to make computers do things at which at the moment, people are better."

Acting Rationally

"Computational Intelligence" is the study of the design of intelligent agents"

"AI is concerned with intelligent behavior in artifacts".

fig: 1(a) ⇒ Some definitions of AI organized into four categories.



⇒ machine give the answer like sum. Run.

Acting Humanly - The Turing Test approach

⇒ The Turing test, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator, after posing some written Qn's, cannot tell whether the written responses come from a person or from a computer. The computer would need to possess the following capabilities.

- 1) Natural language processing ⇒ to enable it to communicate successfully in English.
- 2) Knowledge Representation ⇒ to store what it knows (or hears).
- 3) automated reasoning ⇒ To use the stored information to answer questions and to draw new conclusions.
- 4) Machine learning ⇒ To adapt to new circumstances and to detect and extrapolate patterns.

Turing test deliberately avoided direct physical interaction b/w the interrogator and the computer. Because physical simulation of a person is unnecessary for intelligence. However, the so-called 'Total Turing test' includes a video signal so that the interrogator to pass physical objects, 'Through the hatch'; To pass the total Turing test the computer will need,

* Computer vision - to perceive objects

* robotics - to manipulate objects and move about.

Thinking humanly : The cognitive modeling approach

→ Some way of determining how human think

There are three ways to do this : Through introspection

Trying to catch one own thoughts as they go by.

Through psychobiological experiments - observing the person in action;

Through brain imaging → Observing ^{the} brain in action.

Once we have "sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program. If the program's input-output behavior matches corresponding human behavior, that is evidence that some of the program's machines mechanism could also be operating in humans. For example, Allen

Newell and Herbert Simon, who developed Gps, the 'General problem Solver', were not content merely to have their program solve problems correctly. They were more concerned with - comparing the trace of its reasoning steps - to traces of human subjects solving the same problem. The interdisciplinary field of

Cognitive science brings together computer models from AI, and experimental techniques from psychology to construct precise and

-testable theories of the human mind.

In the early days of AI there was often confusion b/w the approach; an author would argue that an algm performs well on a task and that it is therefore a good model of human performance, or vice versa. Modern authors separate the two kinds of claims, The two fields continue to fertilize each other, most notably in computer vision, which incorporate neurophysiological evidence in to computational models.

Thinking Rationally: The "laws of Thoughts" approach

→ The Greek philosopher Aristotle was one of the first attempt to ~~code~~ codify a right thinking that is, refutable reasoning processes.

→ These laws thought were supposed to govern the operation of the mind; their study initiated the field called logic.

→ There are two main obstacles to this approach -
First it is not easy to take informal knowledge and state it in the formal terms required by logical notation, particularly when the knowledge is less than 100% certain. &

Second → There is a big diff b/w solving a problem "in principle" and solving it

in practice. Even problems with just a few hundred facts can exhaust the computational resources of any computer unless it has some guidance as to which reasoning steps to try first. Although both of these obstacles apply to any attempt to build computational reasoning systems, they appeared first in the logical tradition.

Acting Rationally: The Rational agent approach

⇒ An agent is just something that acts. Of course all computer programs do something, but computer agents are expected to do more: Operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals. A rational agent is one that acts so as to achieve the best outcome or when there is uncertainty, the best expected outcome.

In the "laws of thought" approach to AI, the emphasis was on correct inferences. Making correct inference is sometimes part of being a rational agent. Because one way to act rationally is to reason logically to the conclusion that a given action will achieve one's goals and then to act on that conclusion. On the other hand, correct inference is not all of rationality; in some situations there is no provably correct thing to do, but something must still be done. There are also

ways of acting rationally that cannot be said to involve inference. For example, recoiling from a hot stove is a reflex action that is usually more successful than a slower action taken after careful deliberation.

All the skills needed for the Turing test also allow an agent to act rationally. Knowledge representation and ~~some~~ reasoning enable agents to reach good decisions. We need to be able to generate comprehension sentences in natural language to get by in a complex society. We need learning not only for erudition, but also because it improves our ability to generate effective behavior.

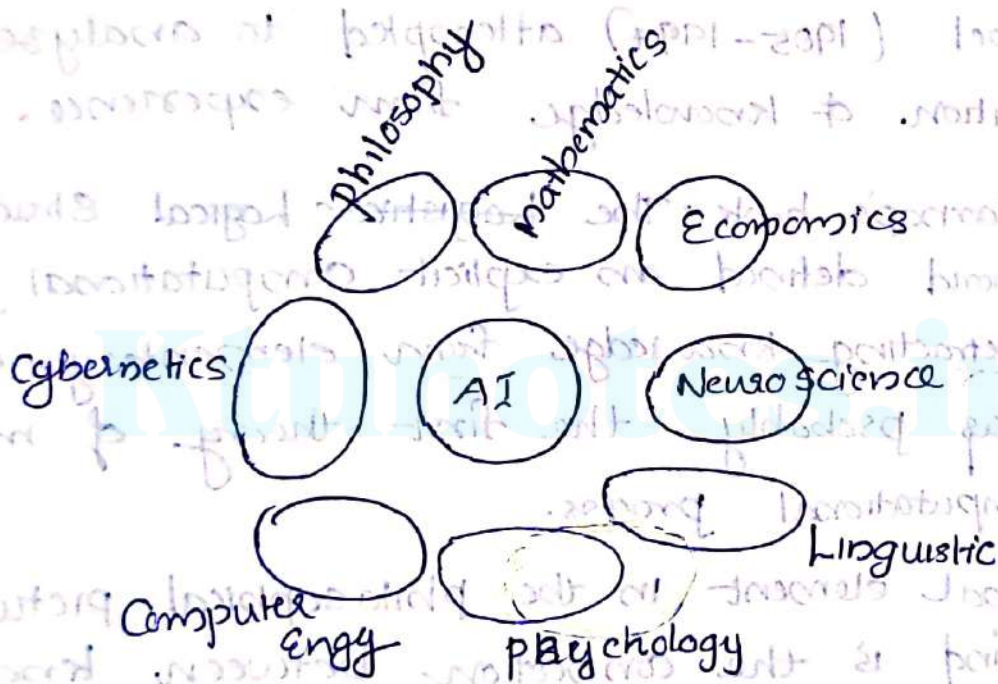
The rational agent approach has two advantages over the other approaches: first, it is more general than the "laws of thought" approach because correct inference is just one of several possible mechanisms for achieving rationality. Second, it is more amenable to scientific development than are approaches based on human behavior or human thought.

The standard of rationality is mathematically well defined and completely general, and can be "unpacked" to generate agent designs that provably achieve it. Human behavior, on the other hand, is well adapted for one specific environment and it is defined by well, the sum total of all the things that human do.

→ we will see before too long that achieving perfect rationality always doing the right thing is not feasible in complicated environment.

The computational demands are just extended. We will adopt the working hypothesis that rationality is a good starting point for analysis. It simplifies the problem and provides the appropriate setting for most of the foundational material in the field.

The Foundation of Artificial Intelligence



Philosophy (basic building blocks)

→ where does knowledge come from?

→ How does the mind arise from a physical brain

* Aristotle (384 - 322 B.C): was the first to formulate a precise set of laws governing the rational part of the mind.
 ↓ is a process of logical actions.

* Thomas Hobbes (1588 - 1679) proposed that reasoning was like numerical computation, that "we add

and subtract in our silent thoughts.

* René Descartes (1596 - 1650): Developed dualistic theory of mind and matter. Descartes attempted to demonstrate the existence of god and the distinction b/w the human soul and body.

* The empiricism ^{from experience} movement, starting with - Francis Bacon's (1561 - 1626)

* The confirmation theory of Carnap and Carl Hempel (1905 - 1997) attempted to analyze the acquisition of knowledge from experience.

* ~~For~~ Carnap's book 'The ~~logistics~~ logical structure of the world defined an explicit computational procedure for extracting knowledge from elementary experience. It was probably the first theory of mind as a computational process.

* The final element in the philosophical picture of the mind is the connection between knowledge and action. This question is vital to AI because intelligence requires action as well as reasoning.

Mathematics

→ What are the formal rules to draw valid conclusions

→ What can be computed.

* George Boole (1815 - 1864), who worked out the details of propositional or Boolean logic (Boole 1847)

* In 1879 - Gottlob Frege (1848 - 1925) extended Boole's logic to include objects and relations, creating the first order logic that is used today.

* The first nontrivial algorithm is thought to be Euclid's algorithm for computing greatest common divisors.

* Besides logic and Computation, the third great contribution of mathematics to AI is the probability theory of probability. The Italian Girolamo Cardano (1501 - 1576) first framed the idea of probability describing it in terms of the possible outcomes of gambling events.

* Thomas Bayes (1702 - 1761) proposed a rule for updating probabilities in the light of new evidence. Bayes rule underlies most modern approaches to uncertain reasoning in AI systems.

Economics

→ Productivity, growth, inequality, market power.

→ From worst outcome choose best one

→ How should we make decisions to maximize payoff

→ How should we do this when the payoff may be far in the future?

* The science of economics got its start in 1776,

Smith was the first to treat it as a science, using the idea that economics can be thought of as consisting of individual agents maximizing their own economic well-being.

* Decision theory, which combines probability theory with utility theory, provides a formal

and complete frameworks for decision (economics or otherwise) made under uncertainty.

- * Von Neuman and Morgenstern's development of game theory included the surprising result that, for some games, a rational agent should adopt policies that are randomized. Unlike decision theory, game theory does not offer an unambiguous prescription for selecting actions.

Neuroscience

→ How do brain process information?

- * Neuroscience is the study of the nervous system particularly the brain.

- * 335 B.C Aristotle wrote "of all the animals, man has the largest brain in proportion to his size."

- * Nicolas Rashevsky (1936, 1938) was the first to apply mathematical models to the study of the nervous system.

- * The measurement of intact brain activity began in 1929 with the invention by Hans Berger of the electroencephalograph (EEG)
↳ To measure brain activity

- * The recent development of functional magnetic resonance imaging (fMRI) is giving neuroscientists unprecedentedly detailed images of brain activity, enabling measurements that corresponds in interesting ways to ongoing cognitive processes

Psychology

→ How do humans and animals think and act?

- * Behaviourism movement, led by John Watson (1878-1958). Behaviorists insisted on studying only objective measures of the percepts (stimulus) given to an animal and its resulting actions (responses). Behaviorism discovered a lot about rats and pigeons but had less success at understanding human.

- * Cognitive psychology views the brain as an information processing device.

- * Common view among psychologists that a cognitive theory should be like a computer program. It should describe a detailed information processing mechanism whereby some cognitive function might be implemented.

Computer Engineering

→ How can we build an efficient computer?

- * The first operational computer was the electro-mechanical ^{code breaking machine} ~~Heath Robinson~~, built in 1940 by Alan Turing team for a single purpose: deciphering German messages.

- * The first operational programmable computer was the Z-3 the invention of Konrad Zuse in Germany.
 Z-3 - German electro-mechanical computer designed by Zuse in 1941

- * The first electronic computer the ABC, was assembled by John Atanasoff and his student Clifford Berry b/w 1940 & 1942 at Iowa State University.
 ↳ early digital computer

- * The first programmable machine was a loom, devised in 1805 by Joseph Marie Jacquard (1752-1834), that used punched cards to store instructions for the pattern to be woven.

Control theory and cybernetics

→ How can artifacts ^{→ object made by human} operate under their own control?

- * Ktesibios of Alexandria built the first self-controlling machine: a water clock with a regulator that maintained a constant flow rate. This invention changed the definition of what an artifact could do.

- * Modern control theory, especially the branch known as Stochastic Optimal control, has as its goal the design of systems that maximize an objective function over time. This roughly OBJECTIVE FUNCTION matches our view of AI: designing systems that behave optimally.

Linguistics

→ How does language relate to thought?

- * In 1957 B.F. Skinner published Verbal Behavior.

This was a comprehensive detailed account of the behaviorist approach to language learning written by the foremost expert in the field.

- * Noam Chomsky, who had just published a book on his own theory, Syntactic Structures; Chomsky pointed out that the

Behaviorist theory did not address the notion of creativity in language.

- * Modern linguistics and AI then were born at about the same time, and grew up together intersecting in a hybrid field called computational linguistic or natural language processing.

The History of Artificial Intelligence

1943 - 1955 - Creation of AI.

- * 1943 - McCulloch & Pitts: Boolean Circuit model of brain.
- * 1950 - Alan Turing's "computing machinery and Intelligence" early AI programs including Samuel's checkers, Newell & Simon's logic theorist, Gretebters geometry engine.
- * 1950-69 - Early enthusiasm, great expectations.
- * 1956 - Dartmouth workshop, name 'artificial intelligence' officially adopted.
- * 1965 - Robinson's complete algorithm for logical reasoning.
- * 1966-74 - AI discovers computational complexity, Neural network research almost disappears.
- * 1969-79 → early development of logic based systems.
- * 1980-88 → expert systems industry booms.
- * 1988-93 → expert systems industry busts - "AI winter"

1980-95 → neural networks return to popularity.

1988 → Resurgence of probability, general increase in technical depth, Nouvelle AI, A Life of AI's, soft computing

1995 → Agents, Agents everywhere.

2003 → present — Human level AI back on agenda

* pattern recognition, NLP, Speech recognition
humanoid robots, web bots, Machine learning
etc.

McCulloch & Pitts (1943)

* proposed a model of artificial neurons.

* Each neuron is characterized as being "ON" or "OFF".

* Switch, to "On" occurring, in response of to stimulation by sufficient no. of neighbouring neurons.

* The state of a neuron was considered of as "fact" factually equivalent to a proposition.

* Any computable function could be computed by some net of connected neurons.

* All the logical connectives (AND, OR, NOT) could be implemented by simple net structures

* McCulloch and Pitts also suggested that suitably defined nets could learn.

* first Neural net computer (1950)

Dartmouth (1956)

- * 2 month, 10 man study of AI
- * Newell and Simon came up with a reasoning programs, the logic Theorist (LT)
- * The program was able to prove most of the theorems in chap 2 Principia Mathematica.

Early Enthusiasm (1952-1969)

- * General problem Solver — Gips (thinking humanly)
- * Herbert Gelernter (1956) constructed the Geometry theorem prover.
- * Arthur Samuel (1956) wrote a series of programs for checkers (draughts) that eventually learned to play at a strong amateur level!
- * Lisp (1956) by John McCarthy.
↳ Second oldest high level programming lang.
- * Knowledge Based System (1969-1979)

Dendral

- * Was an early (probably first) Expert-system developed in 1965 at Stanford University, California.
- * It was a chemical analysis expert system and used Spectrographic data for its analysis
- * Later it was extensively used in academia and industry as the results were as good as any human chemical expert.

Mcgin

- * It too, was an early ES developed in 1972 developed at Stanford.

↳ Human expert decision ability applied machine

- * It diagnosed patients on the basis of symptoms and blood test results.
- * Recommended the course of treatment too.
- * The results were as good as any good human Specialist.

State of the art

- * Deep Blue defeated the reigning world chess-champion Garry Kasparov in 1997.
- * No hands across America driving autonomously 98% of the time from Pittsburgh to San Diego.
- * During the 1991 Gulf war, US forces deployed an AI logistics planning and scheduling program that involved up to 50,000 vehicles, cargo, and people.
- * In 2011 IBM's Watson defeated 2 champions of famous US TV game Jeopardy.
- * NASA's on board autonomous planning program - Controlled the scheduling of operations for a space craft.
- * Proverb solve crossword puzzle better than most humans.

Natural Language processing

* Speech technologies

→ Automatic speech recognition (ASR)

→ Text to speech synthesis (TTS)

→ Dialog systems

* Language processing Technologies:

→ Machine Translation

→ Information Extraction

→ Information Retrieval

→ Text classification, Spam filtering

Application of AI

① Finance

* Bank use AI to:

- organize operations (transactions debit, credit)
- Invest in stocks (stock marketing)
- Manage properties

* Financial institutions have used AI to detect changes (or) claims (downsizing)

② Music

(songs)
* composition, performance, music theory, sound processing and are some of the major areas on which research in music and artificial intelligence are focusing.

③ Transportation

Fuzzy logic controllers have been developed for automatic gearboxes in automobiles.

For eg: Audi TT

VW Toureg

VW Caravel

These features the DSP transmissions which utilizes fuzzy logic.

④ Toys and Games :-

AI has also been applied to video games, for eg: video game bots, which are designed to stand in an opponents where humans - are't available (as desired).

⑤ Hospitals

A medical clinic can use AI system to

- * organize be Schedules

- * Make a staff rotation.

- * provide Medical Information and there

are so many more.

- * Heart Sound analysis.

- * Computer aided Interpretation medical images

Eg: Helps in detection of Tumour.

⑥ Expert system:

- * It is computer based AI which replicates human problem resolving capabilities

- * Under their user has to provide an problem through use interface and interference engine is just like a search engine

⑦ Robotics

⑧ Speech Recognition

⑨ E-commerce

⑩ Computer Vision.

Intelligent agents.

Agents and Environments

- * Agent: An agent is anything that can be viewed as:
- perceiving its environment through sensors
 - it acting upon that environment through actuators.

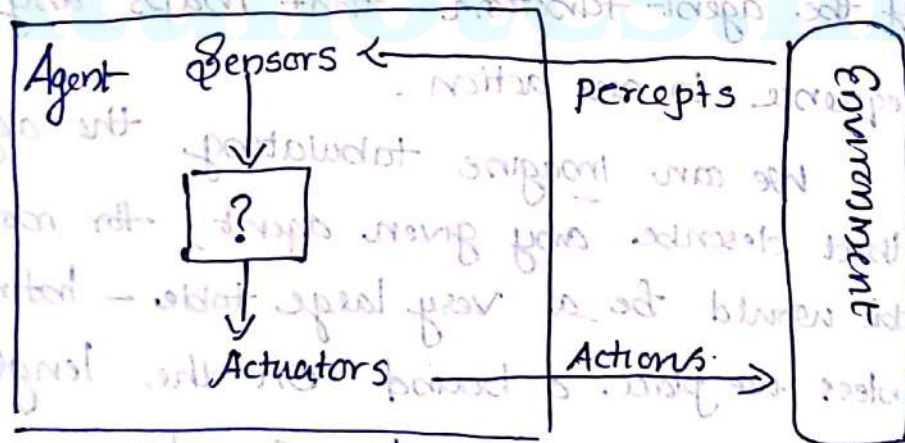


Fig 1(b): Agent interact with environments through sensors and actuators.

→ A human agent has eyes, ears and other organs for sensors and hands, legs, vocal tract, and so on for actuators. A robotic agent might have cameras and infrared range finder for sensors and various motors for actuators.

→ A S/W agent receives keystrokes, file contents, and n/w packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending n/w packet.

→ An percept ⇒ To refer to the agent's perceptual inputs at any given instant.

→ An agent perceptual sequence is the complete history of everything the agent has ever perceived.

"In general, an agent's choice of action at any given instants can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived.

→ Agent function → An agent's behavior is described by the agent function that maps any given percept sequence to an action.

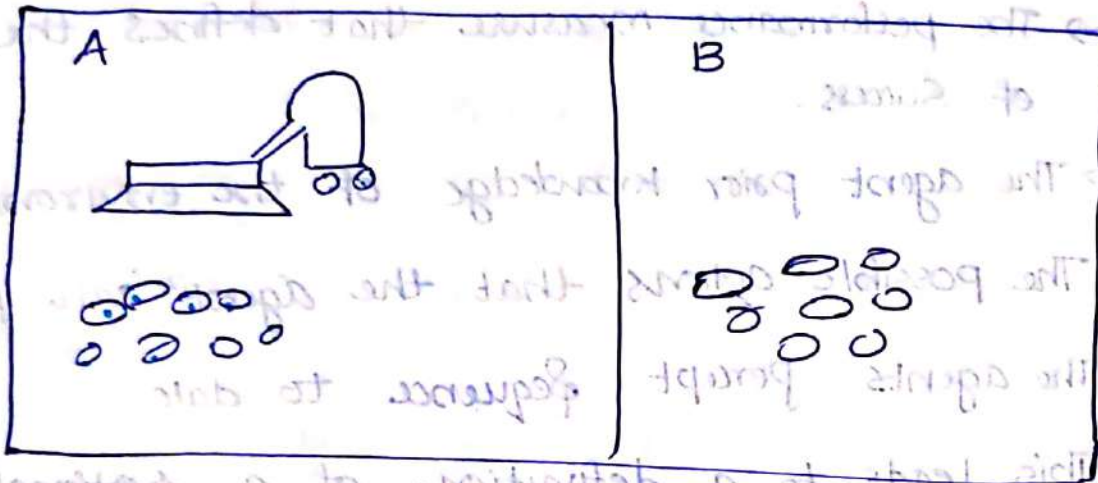
We can imagine tabulating the agent function that describe any given agent; for most agents, this would be a very large table - infinite, in fact unless we place a bound on the length of percept sequences we want to consider.

The table is of course, an external characterization of the agent. Internally, the agent function for an artificial agent will be implemented by an agent program.

It is important to keep these two ideas distinct. The agent function is an abstract mathematical description; the agent

program is a concrete implementation, running with some physical system.

→ To illustrate these idea we use a very simple example - The vacuum cleaner -



* Percepts: Location and Contents eg: (A Dirty)

* Actions: Left, Right, Suck, Noop

* Agent function: mapping from percepts to actions.

Percept	Action
(A, clean)	Right
(A, dirty)	Suck
(B, clean)	Left
(B, dirty)	Suck.

* Good Behavior : The concept of Rationality

Rational agent { "For each possible percept-sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever build-in knowledge the agent has"

Rationality

* Rationality is relative to a performance measure.

* Judge rationality based on

- The performance measure that defines the criterion of success.
- The agent prior knowledge of the environment.
- The possible actions that the agent can perform.
- The agent's percept sequence to date.

This leads to a definition of a rational agent.

"For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has."

Omniscience, learning and autonomy

We need to be careful to distinguish b/w rationality and omniscience. An omniscient agent knows

the actual outcomes of its actions and can act accordingly; but omniscience is impossible in reality. Consider the following example. I am walking along the Champs Elysees one day and I see an old friend across the street. There is no traffic nearby and I am not otherwise engaged, so being rational, I start to cross the street. Meanwhile at 33,000 feet, a cargo

door falls off a passing airliner, and before I make it to the other side of the street lamp flattened. Was it irrational to cross the street. It is unlikely that my obituary would read "Idiot attempts to cross street."

This example shows that rationality is not the same as perfection. Rationality maximize expected performance, while perfection maximize actual performance. Retreating from a requirement of perfection is not just a question of being fair to agents. The point is that if we expect an agent to do what turns out to be the best action after the fact, it will be impossible to design an agent to fulfill this specification - unless we improve the performance of crystal ball or time machine.

Information gathering is an important part of rationality and is covered in depth.

"Doing actions in order to modify future percepts -
↓
called Information gathering."

example of If gathering is provided by the explorer that must be undertaken by a vacuum-cleaning agent in an initially unknown environment.

→ Our definition requires a rational agent not only to gather information but also learn as much as possible from what it perceives.

→ The agent's initial configuration could reflect some prior knowledge of the environment, but as the agent

gains experiences this may be some prior knowledge of the environment, but as the agent gains experiences this may be modified and augmented. There are extreme cases in which the environment is completely known a priori. In such cases, the agent need not perceive (or learn); it simply acts correctly. Of course such agents are fragile.

Consider the lowly dung beetle. After digging its nest and laying its eggs, it fetches a ball of dung from a nearby heap to plug the entrance.

→ To extend that an agent relies on the prior knowledge of its designer rather than on its own percepts, we say that the agent lacks autonomy.

A rational agent should be autonomous; it should learn what it can compensate for partial (or incorrect) prior knowledge. For example, a vacuum-cleaning agent that learns to foresee where and when additional dirt will appear will do better than one that does not.

As a practical matter, one seldom requires complete autonomy from the start, when the agent has had little or no experience; it would have to

act randomly unless the designer gave some assistance. So just as evolution provides animals with enough built-in reflexes to survive long enough to learn for themselves, it would be reasonable to provide an artificial intelligent agent with some initial knowledge as well as

After sufficient experiences of its environment, the behavior of a rational agent can become effectively independent of its prior knowledge. Hence the incorporation of learning allows one to design a single rational agent that will succeed in a vast variety of environment.

The Nature of Environment

→ Task environments which are essentially the "problem" to which rational agents are the "solution". We begin by showing how to specify a task environment, illustrating the process with a number of examples. We then show that task environments come in a variety of flavors. The flavor of the task environment directly affects the appropriate design for the agent program.

Specifying the task environment

In rationality of the simple vacuum cleaner agent we had to specify the performance measure, the environment and the agent's actuators and sensors. We group all these under the heading of task environment.

PEAS → (Performance, Environment, Actuators, Sensors)

→ In designing an agent the first step must always be specified the task environment.

→ The vacuum world was a simple example; let us

Consider a more complex problem; an automated

- taxi driver; we should point out, before the reader become alarmed, that a fully automated is currently somewhat behind the capabilities -

of existing technology. The full driving task is extremely open-ended. There is no limit to the novel combinations of circumstances that can arise - another reason we choose it as a focus for discussions.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal comfortable trip maximize profit	Roads, other traffic, pedestrians customers	Steering accelerator brake, signal horn, display	Cameras, sonar, speedometer keyboard

First performance Measure - to which we would like our automated driver to aspire? Desirable Qualities include getting to the correct destination, minimizing fuel consumption and wear and tear, ^{max} minimizing safety and passenger comfort; maximizing profit. Obviously some of these goals conflict, so tradeoffs will be required.

Next what is the driving environment that the taxi will face? Any taxi driver must deal with a variety of roads, ranging from rural lanes and urban alleys to 12 lane free ways. The roads contain other traffic, pedestrians, stray animals, road works, police cars, puddles, and potholes. The taxi must also interact with potential and actual passengers. There are also some optional choices. The taxi must need to operate in Southern California, where snow is seldom

a problem, or in Alaska, where it seldom is not it could always be driving on the right, or we might want it to be flexible enough to drive on the left when in Britain or Japan. Obviously, the more restricted the environment, the easier the design problem.

The actuators for an automated taxi include those available to a human driver; the control over the engine through the accelerator and control over steering and braking. In addition, it will need output to a display screen or voice synthesizer to talk back to the passengers and perhaps some way to communicate with other vehicles politely or otherwise.

The basic sensors for the taxi will include one or more controllable video cameras so that it can see the road; it might augment these with infrared or sonar sensors to detect distances to other cars and obstacles. To avoid speeding tickets, the taxi should have a speedometer, and to control the vehicle properly, especially on curves, it should have an accelerometer. To determine the mechanical state of the vehicle, it will need the usual array of engine, fuel, and electrical system sensors.

Like many human drivers, it might want a global positioning system (GPS) so that it does not get lost. Finally, it will need a keyboard or microphone for the passenger to request a destination.

→ Software Agent $\Rightarrow$ Some sw exist in web, unlimited domains. Imagine a software website Operator designed to scan internet news sources and show the interesting items to its users, while selling - advertising space to generate revenue. To do well that Operator will need some natural lang processing-abilities, it will need to learn what each user and advertiser is interested in, and it will need to change its plan dynamically.

The Structure of Intelligent Agents

Agent's Structure can be viewed as,

* Agent = Architecture + Agent program.

* Architecture = The Machinery that an agent executes on.

* Agent program = An implementation of an agent-function.

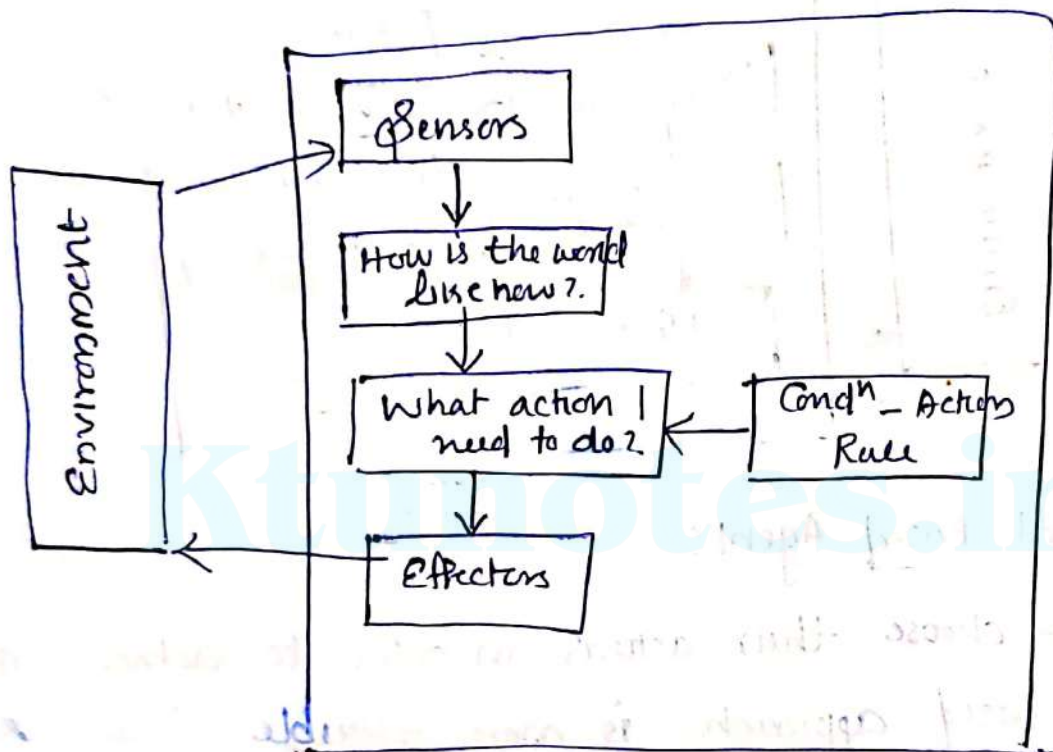
Simple Reflex Agents

* They choose actions only based on the current percept.

* They are rational only if a correct decision is made only on the basis of current percept.

* Their environment is completely observable.

Condition action Rule - It is a rule that maps a state (condition) to an action.



Model Based Reflex Agents

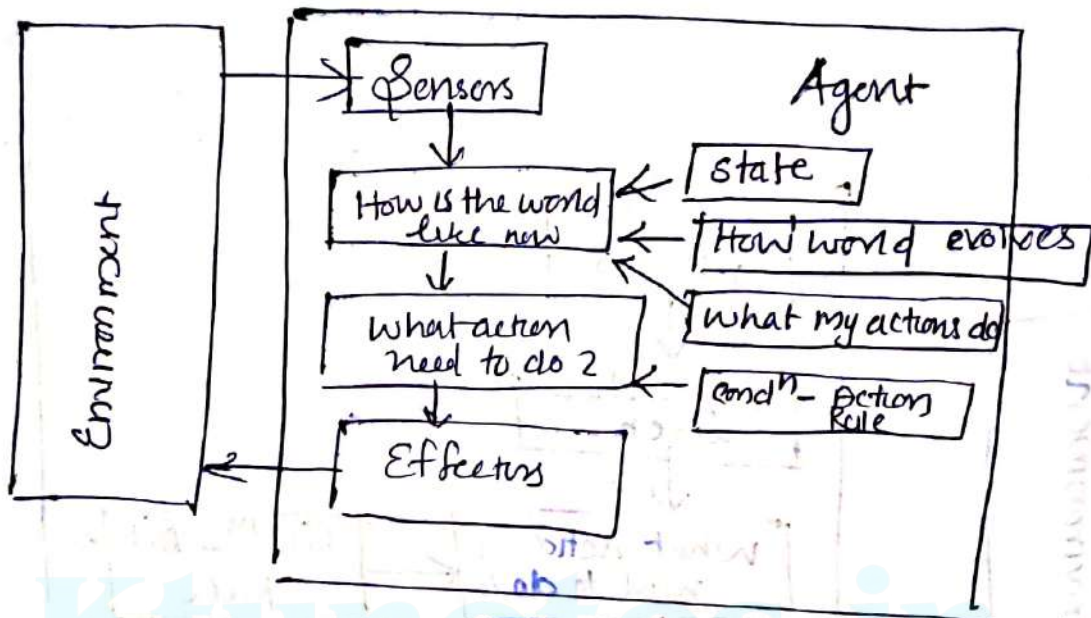
They use a model of the world to choose their actions. They maintain an internal state.

Model $\Rightarrow$ knowledge about "how the things happen in the world."

Internal State $\Rightarrow$ It is a sepⁿ of unobserved aspects of current state depending on percept history.

updating the state requires the information about -

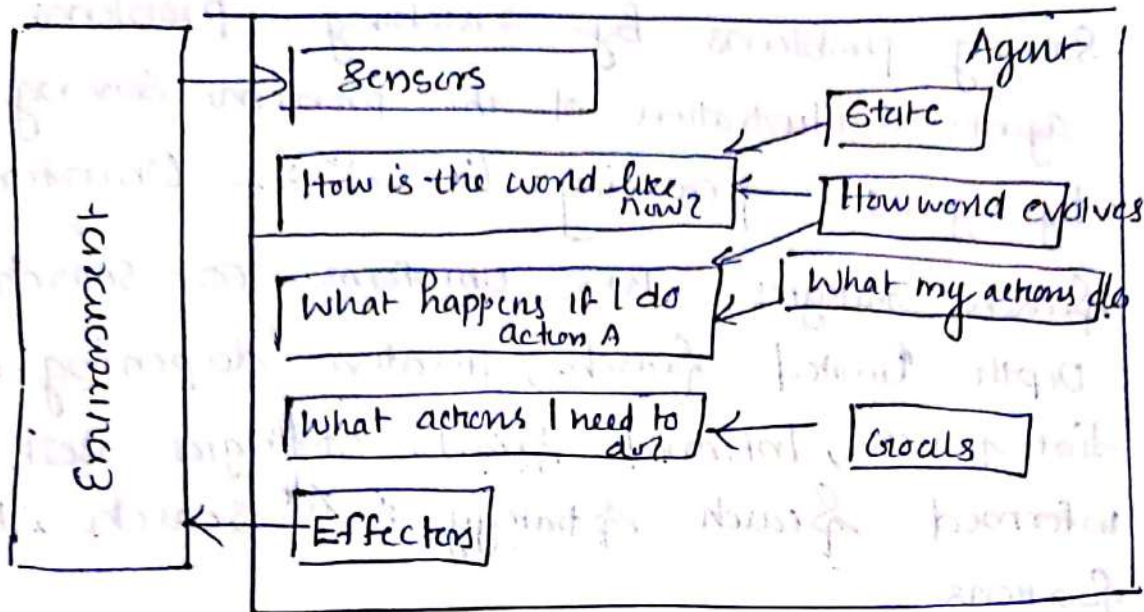
- * How the world evolves
- * How the agents actions affect the world



Goal Based Agents

They choose their actions in order to achieve goals. Goal based approach is more flexible than reflex agent since the knowledge supporting a decision is explicitly modeled, thereby allowing for modifications.

Goal :- It is the description of desirable situation

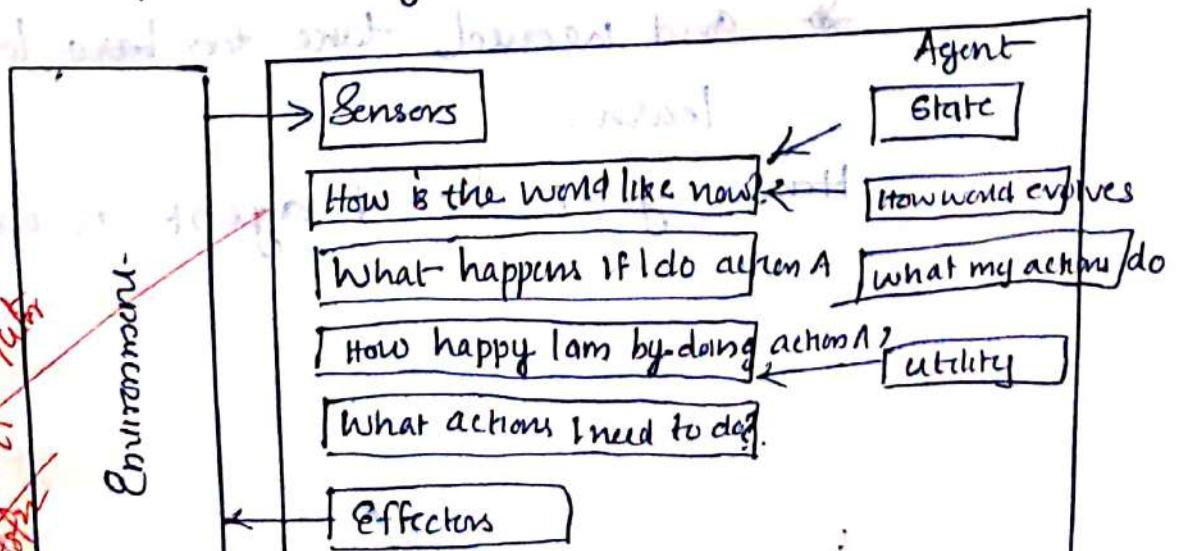


Utility Based Agents

They choose actions based on preference (utility) for each state.

Goals are Inadequate when -

- * There are conflicting goals out of which only few can be achieved.
- * Goals have some uncertainty of being achieved and you need to weigh likelihood of success against the importance of a goal.



Had
Argument
to
show
that